Dental Booking Platform

Development Phases Roadmap — Team Reference

Product: Multi-tenant dental booking SaaS (AWS)
Date: May 2026
Audience: Engineering, product, stakeholders
Estimated timeline: 10–14 weeks to full platform with billing
Repository (planned): `dental-booking-platform`

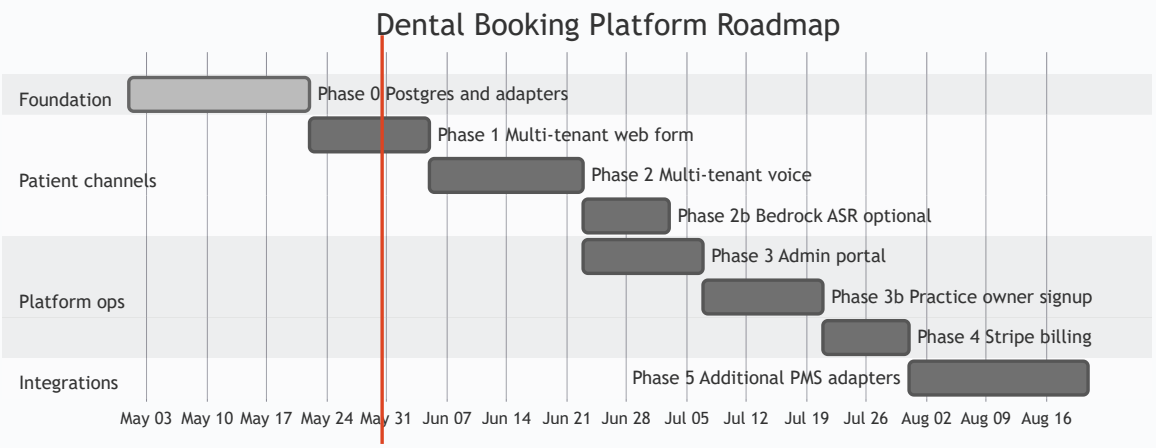
1. Product model (agreed)

Topic	Decision
Who registers	**Practice owners only** — not patients
Users per practice	**One admin user** at signup; bots handle patient-facing tasks
Multi-tenant	Each **clinic = one tenant**; many clinics on one platform
Voice (default)	Amazon Connect + Lex bots (booking, admin, up to 8 per tenant)
Web booking form	**Optional add-on** — `features.webForm: false` at signup; enabled on request
PMS integration	**HTTP APIs** via `PmsAdapter` — Oryx first; no copy of Oryx internal DB schema
GoHighLevel	**Out of scope** — not in AWS product roadmap
Cloud	**AWS only** — ECS, Connect, Lex, RDS, S3, Secrets Manager, Bedrock (optional)

> **Oryx PMS schema reference** (e.g. industry model with patients/appointments tables) is useful for understanding dental data — *not* required to run our platform. We store **tenant config** only; booking uses Oryx MyChart APIs.

2. Phase overview

Phase	Duration	Status	One-line outcome
0 Foundation	2–3 weeks	Implemented	Postgres, tenants, Oryx adapter, internal APIs
1 Web form	~2 weeks	Next	`/book/[slug]`, branding, webForm gate
2 Voice multi-tenant	2–3 weeks	Planned	DID → tenant+bot; Lex uses platform config
2b Bedrock ASR	1–2 weeks	Optional	Name/email correction via LLM
3 Admin portal	~2 weeks	Planned	Operator tools: tenants, bots, phones, assets
3b Practice signup	2–3 weeks	Planned	Self-service registration + admin login
4 Stripe billing	1–2 weeks	Planned	Subscriptions, feature sync, suspend
5 More PMS	Ongoing	Planned	Open Dental, Dentrix, custom API, RPA


Dental Booking Platform Roadmap
Foundation
Patient channels
Platform ops
Integrations
Phase 0 Postgres and adapters
Phase 1 Multi-tenant web form
Phase 2 Multi-tenant voice
Phase 2b Bedrock ASR optional
Phase 3 Admin portal
Phase 3b Practice owner signup
Phase 4 Stripe billing
Phase 5 Additional PMS adapters
May 03
May 10
May 17
May 24
May 31
Jun 07
Jun 14
Jun 21
Jun 28
Jul 05
Jul 12
Jul 19
Jul 26
Aug 02
Aug 09
Aug 16

3. Phase details

Phase 0 — Foundation Implemented

2–3 weeks · Prerequisite for all phases

Goal: Multi-tenant data model and shared booking engine; Smile Squad as tenant #1.

Deliverables:

- Postgres + Prisma (local Docker; RDS via Terraform in follow-up PR)
- Tables: `tenants`, `tenant_pms_config`, `tenant_bots`, `tenant_phone_numbers`, `subscriptions` (stub)
- `TenantService`, `PmsAdapter` + `OryxAdapter`, per-tenant operatory rules JSON
- APIs: `GET /api/internal/resolve-phone`, `/api/t/[slug]/public-config|availability|book`
- Legacy routes preserved: `/api/availability`, `/api/web/book`, `POST /api/book` (secured with `CONNECT_WEBHOOK_SECRET`)
- Seed tenant `smilesquad` (Oryx realm `smilesquadpd`)
- Lex handler unchanged except Phase 2 TODO — still calls platform HTTP APIs

Not in Phase 0: Slug web UI, Lex tenant lookup, Stripe, self-service signup, admin portal, Bedrock.

Phase 1 — Multi-tenant web booking form Next

~2 weeks

Goal: Public web booking per clinic slug; web form only when add-on enabled.

Deliverables:

- `/book/[slug]` — dynamic branding from `tenants.branding`
- Redirect `/book` → `/book/smilesquad` (backward compat)
- Wire UI to `/api/t/[slug]/availability` and `/api/t/[slug]/book`
- Feature gate: **403** if `features.webForm` is false
- Per-tenant access code (`web_form_access_code`)
- Default new tenants: `webForm: false` until ops/Stripe enables add-on

Who uses it: Patients (no login) book into tenant's Oryx realm via web.

Phase 2 — Multi-tenant voice bots Planned

2–3 weeks

Goal: One platform serves many clinics on voice; DID selects tenant + bot.

Deliverables:

- Lex Lambda calls `resolve-phone` at call start (DID → tenant, bot, operatory rules, prompts)
- Replace hardcoded "Smile Squad" copy with tenant/bot `prompts_config`
- Remove duplicated operatory logic from Lambda — use platform rules
- Support up to **8 bots per tenant** (booking, admin, FAQ, etc.)
- Connect: map each DID to `tenant_phone_numbers`
- `POST /api/book` with `X-Tenant-Slug` where needed

Who uses it: Patients call practice number; bots book via same Oryx APIs as web.

Phase 2b — Bedrock ASR correction Optional

1–2 weeks (after Phase 2)

Goal: Improve voice capture of names, email, and correction turns.

- `POST /api/internal/normalize-slot` (platform secret)
- Amazon Bedrock for structured extraction; mandatory read-back before book
- Per-tenant feature flag; heavier use on admin/FAQ bot
- Booking APIs remain deterministic (no LLM in final book call)

Phase 3 — Admin portal (operator) Planned

~2 weeks

Goal: Internal/ops team manages tenants without SQL.

- `/admin` — tenant CRUD, PMS config, operatory rules editor
- Bot registry (create/disable, Lex IDs, Lambda ARNs, prompts JSON)
- Phone → bot mapping
- Enable/disable `features.voice` and `features.webForm` (web form add-on)
- S3 upload for logos / hero images
- Booking health dashboard (success/fail per tenant)

Onboarding v1: Operator creates tenant (manual) — no public signup yet.

Phase 3b — Practice owner self-service Planned

2–3 weeks (after Phase 3 or with Phase 4)

Goal: Practice owner registers, picks PMS (Oryx first), gets one admin login.

- New table: `platform_users` (email, password, role, `tenant_id`)
- Registration flow: practice info → select PMS type → provision tenant
- Default: `voice: true`, `webForm: false`
- "Request web form" workflow → ops approves → flip feature + access code
- Practice admin dashboard (limited): view status, request add-ons, basic config

Not included: Patient accounts, staff RBAC beyond one admin (later).

Phase 4 — Stripe subscriptions Planned

1–2 weeks

- Stripe products: voice plans, web form add-on
- `POST /api/webhooks/stripe` — update `subscriptions`, sync `tenants.features`
- Suspend tenant on `past_due` / `canceled` (403 on book + voice routing)
- Optional: self-serve checkout linked to Phase 3b signup

Phase 5 — Additional PMS adapters Ongoing

Ongoing after Phase 4

- `OpenDentalAdapter`, `DentrixAdapter`, `CustomApiAdapter`
- Same `PmsAdapter` interface — UI and Lex unchanged
- RPA/SQS workers for PMS without public APIs (future; not FastAPI rewrite in current plan)

4. What we store vs what Oryx stores

Our Postgres (platform)	Oryx / PMS (via API)
Tenant slug, branding, features, bots, DIDs	Patients, charts, appointments
`realm`, operatory rules, adapter config	Providers, operatories, schedule slots
Stripe subscription status	Clinical, billing, insurance (not mirrored)
Platform admin users (Phase 3b)	Practice staff in PMS (`staff_users` in PMS model only)

5. Channels end-state

Channel	Entry	Phase
Web booking	`/book/{slug}`	Phase 1 (APIs ready in Phase 0)
Voice booking	Connect DID → Lex → `/api/book`	Phase 2
Practice admin	`/admin` or signup portal	Phase 3 / 3b
Internal routing	`/api/internal/resolve-phone`	Phase 0 ✓

6. Infrastructure (by phase)

Component	Phase
Docker Postgres (local)	0 ✓
RDS Postgres + Terraform	Follow-up PR after 0
ECS Fargate + ALB (existing)	Ongoing
S3 tenant assets	3
ACM + Route53 TLS	3+
Bedrock	2b (optional)

7. Success criteria (release-ready platform)

- Second clinic onboarded with **DB config only** (no app redeploy)

- `/book/{slug}` shows correct branding; books correct Oryx realm
- Call to Clinic B's DID uses B's bot + PMS config
- Voice-only tenant gets **403** on web form
- Web form add-on enables `webForm` + access code without redeploy
- Practice owner can register (Phase 3b) and manage one admin account
- Stripe cancellation suspends tenant within one webhook cycle (Phase 4)
- Second PMS adapter plugs in without UI/Lex flow changes (Phase 5)

8. Explicitly out of scope

- GoHighLevel integration and `tenant_ghl_locations`
- Patient registration on our platform
- Replicating Oryx's 22-table PMS database in our Postgres
- FastAPI rewrite (stay on Next.js full-stack)
- DynamoDB, EventBridge, SQS RPA workers (until Phase 5+ path)

> **Phase 0 code location:** `oryx-agent/` today; planned clean repo `dental-booking-platform` (no GHL) for team delivery.
